**Alette Wells**
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## EDUCATION

**Yale University,** New Haven, CT — Expected Graduation: December 2024
*BA in Mathematics, Certificate in French*
*GPA:* 3.98
*Awards:* Phi Beta Kappa (Junior Year, January 2023), Montaigne Prize (1st Place, French Dept., May 2023), The Scott Prize (Honorable Mention, French Dept., May 2023), Nathan Hale Associate (May 2021).

**Palisades Charter High School,** Los Angeles, CA — June 2020
*Awards:* Honors Roll (2016-2020), Prize for Excellence in Mathematics (2020)
*Extracurricular:* Los Angeles Math Circle, UCLA (2013-2020)

## RESEARCH EXPERIENCE

**SUAMI,** *Carnegie Mellon University*, Pittsburgh, PA — Summer 2024
*Project Title*: "Connectedness of Special Points in the Markoff mod p Graphs"
*Mentor*: Elisa Bellah (University of Toronto)

- Studied the connectedness of graphs constructed from the mod p solutions to the Markoff equation; connectedness of a mod p graph is equivalent to the statement that every mod p solution to the Markoff equation has an integer lift
- Devised heuristic lower bounds on the number of points lying in the "cage," a component of the Markoff mod p graph which is always connected
- Identified and proved explicit algebraic conditions guaranteeing that a certain class of Markoff mod p points are connected to the cage.
- Paper in progress; accepted for poster presentation at JMM January 2025

**SUMRY,** *Yale University*, New Haven, CT — Summer 2022
*Project Title*: "Geometry and Dynamics in $H^2$ x $H^2$"
*Mentors*: Or Landesberg and Subhadip Dey (Yale University)

- Examined geodesic dynamics on quotients of $H^2$ x $H^2$. Focused on the quotient by Fuschian groups satisfying specific constraints in order to characterize Benoist's limit cone in this case
- Project culminated in a presentation at the Young Mathematician's Conference (The Ohio State University)

**Polymath Jr. REU,** Remote — Summer 2021
*Project Title*: "Gelfand-Tsetlin Polytopes"
*Mentors*: Ben Brubaker and Andy Hardt (University of Minnesota)

- Studied *colored Gelfand-Tsetlin* patterns; devised rules that the coloring scheme must obey in order to be well-defined, to refine the uncolored GT patterns, and to retain information about the underlying representation theory that these patterns encode
- Characterized properties of colored GT patterns, with the goal of establishing novel proofs for classical results about the generating functions for standard GT patterns

**Accepted to Budapest Semesters in Mathematics** — Spring 2023

- (Could not attend for financial reasons)

## WORK EXPERIENCE

**Undergraduate Learning Assistant, Abstract Algebra** — *Yale University*, Sept. 2024- Dec. 2024

- Ran tutoring sessions and answered students' questions about course material, homework assignments
- Wrote practice problems related to topics covered in class, and organized exam review sessions for students
- Wrote weekly reports on student progress and difficulties, so that the professor could gauge their understanding of course material

**Undergraduate Mathematics Grader** *Yale University*, Sept. 2022- Dec. 2023

- Assessed students' written arguments for problem sets in proof-based courses including linear algebra and complex analysis as well as the theory of Lebesgue measures and Fourier analysis
- Wrote constructive and concise feedback on students' assignments in order to improve their understanding of the course material
- Composed weekly reports on student performance and progress for the professor, in order to ensure greater continuity between material presented in lecture and on problem sets

**Peer Tutor** *Yale University*, Sept. 2021- May 2022

- Tutored Yale students in introductory calculus and macroeconomics courses (MATH 112, ECON 116), helped them to develop quantitative reasoning skills, and prepared them for the rigor of university-level STEM courses
- Advised students with a wide-ranging set of mathematical backgrounds about how to seek academic support, as well as about which QR courses would advance their academic goals

**ONEXYS Coach** *Yale University,* June-July 2021

- Tutored incoming Yale first-years from weaker math backgrounds in algebra and pre-calculus
- Developed weekly curricula, with an emphasis on teaching challenging material through fun games and worksheets, and emphasizing the real-world applications of concepts
- Advised students on how to choose university-level math courses, and helped them develop confidence in their mathematical abilities

**Undergraduate Accessibility Aide** *Yale University, SAS and IT Depts.* June-Aug. 2021

- Edited Yale course materials to ensure compliance with accessibility standards for students with sight- and sound-impairments or learning disabilities
- Tested and reviewed math education software intended for students with disabilities, and advised the program director on the software's capabilities to address student needs

**Undergraduate Technology Assistant** *Yale University,* Jan.-May 2021

- Assisted professors in using Zoom, Canvas and other digital platforms so that they could effectively organize their courses during the COVID-19 pandemic.

## LEADERSHIP EXPERIENCE

**Dimensions,** *Events Coordinator*, New Haven, CT Sep 2022-Sep 2023

- Organized and budgeted social events in order to increase undergraduate participation in Yale's chapter of the Association for Women in Mathematics (AWM)
- Coordinated with female math faculty from various universities in order to organize colloquium talks for undergraduates

**Disability Empowerment for Yale,** *Events Coordinator*, New Haven, CT Aug 2020-Aug 2022

- Organized panel discussions between eminent scholars of disability studies, as well as other events that highlighted the disproportionate impact of COVID-19 measures on individuals with disabilities

**Harbor Fellows,** *Peer advisor* Sep 2024 –

- Meet monthly with a first year female student who is a Harbor fellow and a prospective math or applied math major. Advise her on navigating both mathematics and life at Yale.

## SKILLS AND INTERESTS

- *Computer Skills:* Microsoft office suite, Adobe Creative Cloud, Mathematica, LaTeX, Python (Sage, Manim), learning ML
- *Language Skills:* French, L5
- *Interests:* Waterpolo, trumpet, visiting art and history museums, French-language literature and philosophy. Arts and Crafts.